

SAURABH VISHWAKARMA

+91-9695877472 | sv9327545@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

GenAI developer and dual-degree candidate (IIT Madras + Chandigarh University) with hands-on experience building production-grade AI systems spanning Retrieval-Augmented Generation (RAG) pipelines, LLM API integration, agentic AI workflows, and full MLOps infrastructure. Proven ability to translate ML model performance into quantifiable business impact. Proficient in Python, LangChain, FastAPI, Docker, and vector search technologies. Seeking to contribute to enterprise GenAI application development at scale.

EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Science, Data Science and Applications</i>	2023 – 2027
Chandigarh University <i>Bachelor of Computer Applications (BCA)</i>	2023 – 2026

TECHNICAL SKILLS

Languages & Databases: Python, SQL
GenAI & LLM Frameworks: LangChain, Sentence Transformers, Hugging Face Inference API, Prompt Engineering, RAG Pipelines, LLM API Integration (OpenAI-compatible, Anthropic, HuggingFace)
Agentic AI: Tool Calling, ReAct Pattern, Function Calling, Agentic Workflow Design, LangGraph (Familiar), CrewAI (Familiar)
Vector Search: FAISS, Chroma (Familiar), Semantic Search, Cosine Similarity, Embedding-based Retrieval
ML / AI: PyTorch, Scikit-learn, XGBoost, Cost-Sensitive Learning, SMOTE, NLP, Deep Learning
MLOps & Infrastructure: FastAPI, Docker, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, REST APIs
Cloud & Tools: AWS (Familiar), SageMaker (Concepts), Azure OpenAI (Concepts), Git, Plotly, Matplotlib

PROJECTS

Credit Card Fraud Detection System | *Python, XGBoost, FastAPI, Docker, MLflow, Prometheus, GitHub Actions*

- Built an end-to-end ML pipeline for fraud classification achieving **ROC-AUC of 0.9785** and **PR-AUC of 0.6922** on a highly imbalanced dataset (<1% fraud rate) using XGBoost with SMOTE oversampling.
- Designed a **cost-sensitive evaluation framework** with asymmetric business costs (\$10 false positive / \$200 false negative); threshold optimization reduced expected cost from \$55,360 to \$5,200 — a **90% reduction in projected fraud losses**.
- Conducted **segment-level evaluation** across ATM (ROC-AUC: 0.9861) and online (ROC-AUC: 0.9426) transaction types, identifying channel-specific model behaviour and enabling targeted risk thresholds.
- Deployed a **production-grade MLOps stack**: FastAPI inference endpoint, Docker containerization, MLflow model registry with versioning, Prometheus + Grafana monitoring dashboards, and GitHub Actions CI/CD pipeline.

ATS Intelligence Engine v2 — GenAI Document Analysis Platform | *Python, LangChain, RAG, Sentence Transformers, FastAPI*

- Designed and deployed an **end-to-end RAG pipeline** for resume-to-JD matching: document ingestion, chunking, embedding via Sentence Transformers (all-MiniLM-L6-v2), cosine similarity retrieval, and grounded LLM response generation.
- Integrated **LLM-powered feedback generation** using LangChain + Hugging Face Inference API with structured ChatPromptTemplates — producing human-readable skill gap analysis and actionable improvement recommendations.
- Architected **dual FastAPI endpoints**: `/score` (sub-second semantic similarity, no LLM) and `/feedback` (full RAG chain with LLM), demonstrating awareness of latency-cost tradeoffs in GenAI product design.
- Deployed as a **single-container Docker application on Hugging Face Spaces**; applied prompt engineering, structured output formatting, and output validation to ensure consistent, reliable LLM responses.

LeafLens — Plant Leaf Super-Resolution | *Python, PyTorch, SRGAN, Deep Learning*

- Designed and trained a **Super-Resolution GAN (SRGAN)** achieving 4× upscale on plant leaf imagery with **SSIM of 0.91** and **PSNR of 31.2 dB**; implemented custom generator and discriminator architectures in raw PyTorch.
- Managed full training loop including perceptual loss functions, adversarial optimization, and gradient clipping — demonstrating deep understanding of generative model internals applicable to modern LLM-adjacent architectures.

GRU Language Model — Built from Scratch | *Python, PyTorch, NLP, Deep Learning*

- Implemented a character-level **Gated Recurrent Unit (GRU) language model entirely in raw PyTorch** — no high-level wrappers — with custom training loop, gradient clipping, backpropagation through time, and text generation pipeline.
- Demonstrates foundational understanding of sequence modelling and neural network internals directly relevant to LLM architecture comprehension and GenAI development.

RELEVANT COURSEWORK

IIT Madras (B.S. Data Science): Machine Learning Foundations, Deep Learning, Statistics for Data Science, Database Management Systems, Python Programming, Linear Algebra, Probability & Statistics

Chandigarh University (BCA): Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Software Engineering, Cloud Computing Fundamentals

ACHIEVEMENTS & LEADERSHIP

Dual Degree: Simultaneously pursuing B.S. Data Science at IIT Madras and BCA at Chandigarh University — demonstrating capacity for high-volume, rigorous academic workload alongside production-level project development.

Chess: Two-time Inter-College Chess Champion — analytical thinking and strategic decision-making under pressure, directly applicable to multi-step agentic reasoning design.

Leadership: Former Lead Vocalist of an award-winning college band — cross-functional team coordination, performance under scrutiny, stakeholder engagement.

ADDITIONAL INFORMATION

GitHub: github.com/saurabh-004 — All 4 projects deployed and publicly accessible

Portfolio: saurabh-004.github.io/Portfolio

Languages: English (Professional), Hindi (Native)

Open to: Full-time roles, relocation (Noida / Pune / Bangalore / Remote)